The Exact Marked Bridge Between the Erdős–Straus Divisor Selector and the S^6 Terminal Presentation

Independent reconstruction note

4 September 2026

1 The exact marked Erdős–Straus–S^6 bridge

Result and scope

For each prime $p \equiv 1 \pmod{12}$, the complete first-half Erdős–Straus divisor selector is in explicit bijection with a normalized, positively oriented, divisor-supported, and shell-marked slice of the terminal parameter lattice occurring in the candidate complex structure on S^6. The coordinate change is integral in both directions. More strongly, the selector relation matrix and the S6 terminal relation matrix have exactly the same row lattice: they differ by a displayed element of $\mathrm{SL}_2(\mathbb{Z})$. Composing the bijection with the shell reconstruction gives positive denominators solving $4/p = 1/x + 1/y + 1/z$, and every positive solution is detected by at least one of the two tagged slices.

The nonlinear condition $u \mid a_j^2$, together with the external labels (p, j, ε), is indispensable. If it is removed, each normalized S6 fibre is an infinite affine line and carries no Erdős–Straus existence information. A particularly natural unmarked S6 point produces precisely the already-solved congruence class $p \equiv 13 \pmod{24}$ and produces no selector state when $p \equiv 1 \pmod{24}$. Thus the bridge is an exact arithmetic equivalence of marked presentations, not a proof of universal selector nonemptiness and not a geometric morphism from the reciprocal cubic to the complex threefold.

1.1 All objects and conventions

Fix throughout a prime

$$p \equiv 1 \pmod{12}, \qquad h = \frac{p-1}{12} \in \mathbb{Z}_{>0},$$

and write

$$L = \frac{p-1}{4} = 3h, \qquad J_p = \{j \in \mathbb{Z} : 0 \le j < L\}.$$

For $j \in J_p$, retain the four integers

$$R_j = 4j + 3, \qquad a_j = 3h + j + 1 = \frac{p + R_j}{4}, \qquad N_j = pa_j, \qquad 4a_j = p + R_j. \tag{1.1}$$

No label in (p, j) will be suppressed. The inequalities

$$3 \le R_j \le p - 2, \qquad \frac{p+3}{4} \le a_j \le \frac{p-1}{2} < p$$

give

$$\gcd(p, a_j) = \gcd(p, R_j) = \gcd(a_j, R_j) = 1. \tag{1.2}$$

Indeed p is prime and $0 < a_j < p$; then $R_j = 4a_j - p$ gives the other two equalities.

Let

$$\mathscr{S}(p) = \left\{(x, y, z) \in \mathbb{Z}^3_{>0} : 4xyz = p(xy + xz + yz)\right\}.$$

The equation is symmetric, but whenever completeness is invoked we choose the ordered representative $x \le y \le z$. Write ES(p) for the assertion $\mathscr{S}(p) \neq \varnothing$.

For the S6 terminal parameters put

$$\lambda(\ell_0, \ell_1, \ell_2) = 12\ell_0 - 4\ell_1 - 3\ell_2 \tag{1.3}$$

and define the abstract terminal group

$$G(\ell_0, \ell_1, \ell_2) = \left\langle c, x, y \;\middle|\; c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \right\rangle. \tag{1.4}$$

After eliminating y, and using the ordered generator basis (x, c), its ordered row-relation matrix is

$$\mathsf{R}(\ell_0, \ell_1, \ell_2) = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det \mathsf{R} = -\lambda(\ell_0, \ell_1, \ell_2). \tag{1.5}$$

The letter p in this note always denotes the Erdős–Straus prime. The signed terminal value is denoted

$$q_\varepsilon = p^\varepsilon \in \{1, p\}, \qquad \varepsilon \in \{0, 1\}.$$

Thus no identification of the external prime p with the terminal value is made in the $\varepsilon = 0$ channel.

For any integral 2×2 matrix A, define its row-relation lattice and row-presented group by

$$\mathrm{Rel}(A) = \{v^{\mathsf{T}} A : v \in \mathbb{Z}^2\} \subseteq \mathbb{Z}^2, \qquad \mathcal{G}(A) = \mathbb{Z}^2 / \mathrm{Rel}(A) = \mathrm{coker}(A^{\mathsf{T}}). \tag{1.6}$$

This convention makes every matrix direction in the bridge explicit.

1.2 The complete first-half shell selector

Lemma 1.1 (first-half denominator)**.** *Let $p \equiv 1 \pmod 4$ be an odd prime. If $(x, y, z) \in \mathscr{S}(p)$ and $x \le y \le z$, then*

$$\frac{p}{4} < x \le \frac{p}{2}.$$

Proof. Positivity of the two reciprocals other than $1/x$ gives $1/x < 4/p$, hence $x > p/4$. Suppose $x > p/2$. Clearing denominators gives

$$4xyz = p(xy + xz + yz),$$

so $p \mid xyz$. If $p \mid x$, then $x, y, z \ge p$ and the reciprocal sum is at most $3/p$. If $p \mid y$, then $y, z \ge p$, while $1/x < 2/p$, so the sum is strictly less than $4/p$. Both cases are impossible. Therefore $z = kp$ for some $k \in \mathbb{Z}_{>0}$, while neither x nor y is divisible by p. Substitution yields

$$(4k - 1)xy = kp(x + y).$$

Reduction modulo p gives $p \mid 4k - 1$. Write $4k - 1 = rp$. Since $p \equiv 1 \pmod 4$, one has $r \equiv 3 \pmod 4$, hence $r \ge 3$. After cancelling p,

$$rxy = k(x + y).$$

As $x \le y$, division by xy gives

$$x \le \frac{2k}{r} = \frac{rp + 1}{2r} = \frac{p}{2} + \frac{1}{2r} < \frac{p + 1}{2}.$$

The integer $x > p/2$ must be at least $(p + 1)/2$, a contradiction. □

For fixed $j \in J_p$, define the ordered residual shell

$$\mathscr{P}_{p,j} = \{(y,z) \in \mathbb{Z}^2_{>0} : R_j yz = N_j(y+z)\} \tag{2.1}$$

and the oriented divisor shell

$$\mathscr{D}_{p,j} = \{d \in \mathbb{Z}_{>0} : d \mid N_j^2,\ d \equiv -N_j \pmod{R_j}\}. \tag{2.2}$$

Proposition 1.2 (oriented shell bijection)**.** *The maps*

$$\begin{aligned} \delta_{p,j} &: \mathscr{P}_{p,j} \longrightarrow \mathscr{D}_{p,j}, & (y,z) &\longmapsto R_j y - N_j, \\ \rho_{p,j} &: \mathscr{D}_{p,j} \longrightarrow \mathscr{P}_{p,j}, & d &\longmapsto \left(\frac{N_j+d}{R_j}, \frac{N_j+N_j^2/d}{R_j}\right) \end{aligned}$$

are mutually inverse. The involution

$$\iota_{N_j}(d) = \frac{N_j^2}{d}$$

corresponds exactly to $(y,z) \mapsto (z,y)$*.*

Proof. Abbreviate $R = R_j$ and $N = N_j$. Equation (2.1) is equivalent to

$$(Ry - N)(Rz - N) = N^2. \tag{2.3}$$

Both factors are positive: for example, division of $Ryz = N(y+z)$ by z gives $Ry = N(1+y/z) > N$, and similarly for $Rz - N$. Therefore $d = Ry - N$ is a positive divisor of N^2 and $d \equiv -N \pmod{R}$.

Conversely, let $d \in \mathscr{D}_{p,j}$ and put $e = N^2/d$. By (1.2), d and N are units modulo R; hence

$$e \equiv N^2(-N)^{-1} \equiv -N \pmod{R}.$$

Thus both components of $\rho_{p,j}(d)$ are positive integers. Their two shifted factors are d and e, so (2.3), and therefore (2.1), holds. The formulas visibly compose to the identity. Replacing d by $e = N^2/d$ exchanges the two reconstructed components. □

For $\varepsilon \in \{0,1\}$, define the fixed-shell selector and the fully marked selector by

$$\mathcal{W}_\varepsilon(p,j) = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + p^\varepsilon\}, \tag{2.4}$$

$$\mathcal{W}_\varepsilon(p) = \{(j,u) : j \in J_p,\ u \in \mathcal{W}_\varepsilon(p,j)\}. \tag{2.5}$$

The tag $\varepsilon = 0$ is the exterior, or E, channel; $\varepsilon = 1$ is the middle, or M, channel. Whenever a disjoint union of the two channels is written below, it means the tagged coproduct

$$\{0\} \times \mathcal{W}_0(p) \ \sqcup\ \{1\} \times \mathcal{W}_1(p).$$

This convention is necessary because the two raw subsets can intersect.

Proposition 1.3 (exact two-channel decomposition)**.** *The maps*

$$\begin{aligned} \iota_E &: \mathcal{W}_0(p,j) \longrightarrow \mathscr{D}_{p,j}, & u &\longmapsto p^2 u, \\ \iota_M &: \mathcal{W}_1(p,j) \longrightarrow \mathscr{D}_{p,j}, & u &\longmapsto pu \end{aligned}$$

are bijections onto the subsets on which the 2-adic valuations are 2 *and* 1*, respectively. The valuation-zero subset is* $\iota_{N_j}(\iota_E(\mathcal{W}_0(p,j)))$*. Consequently*

$$\mathscr{D}_{p,j} = \iota_E(\mathcal{W}_0(p,j)) \ \sqcup\ \iota_M(\mathcal{W}_1(p,j)) \ \sqcup\ \iota_{N_j}(\iota_E(\mathcal{W}_0(p,j))), \tag{2.6}$$

and

$$|\mathscr{D}_{p,j}| = 2|\mathcal{W}_0(p,j)| + |\mathcal{W}_1(p,j)|. \tag{2.7}$$

Proof. Because $p \nmid a_j$, every divisor $d \mid p^2 a_j^2$ has $v_p(d) \in \{0, 1, 2\}$.

If $d = p^2 u$, then $u \mid a_j^2$. The shell congruence is

$$R_j \mid d + N_j = p(pu + a_j).$$

After cancelling the unit p modulo R_j and multiplying by 4, the identity $4a_j = p + R_j$ makes this equivalent to $R_j \mid 4u + 1$. This proves the E-channel assertion in both directions. If $d = pu$, the same calculation gives

$$R_j \mid d + N_j \iff R_j \mid u + a_j \iff R_j \mid 4u + p,$$

which proves the M-channel assertion. Complementation preserves the shell congruence by Proposition 1.2, exchanges valuations 0 and 2, and preserves valuation 1. This proves the disjoint decomposition and its cardinality formula. □

For $u \in \mathcal{W}_\varepsilon(p, j)$, define the two exact quotients

$$T = \frac{4u + p^\varepsilon}{R_j}, \qquad K = \frac{p^{1-\varepsilon} T + 1}{4}. \tag{2.8}$$

Since $R_j \equiv 3 \pmod 4$ and $p^\varepsilon \equiv 1 \pmod 4$, the first equation gives $T \equiv 3 \pmod 4$. Hence $K \in \mathbb{Z}_{>0}$. Direct calculation, without dropping the tag, gives

$$R_j K = a_j + p^{1-\varepsilon} u, \qquad T a_j = u + p^\varepsilon K. \tag{2.9}$$

Indeed, multiplication by 4 changes these identities into

$$\begin{aligned} 4R_j K &= p^{1-\varepsilon}(4u + p^\varepsilon) + R_j = 4p^{1-\varepsilon} u + p + R_j, \\ 4T a_j &= T(p + R_j) = pT + 4u + p^\varepsilon = 4u + 4p^\varepsilon K. \end{aligned}$$

Proposition 1.4 (denominator reconstruction)**.** *The map*

$$\Psi_{p,j,\varepsilon} : \mathcal{W}_\varepsilon(p, j) \longrightarrow \mathscr{S}(p), \qquad u \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u} \right) \tag{2.10}$$

is well-defined. Its two oriented shell factors are

$$R_j(pK) - N_j = p^{2-\varepsilon} u, \qquad R_j \left(\frac{p^\varepsilon a_j K}{u} \right) - N_j = p^\varepsilon \frac{a_j^2}{u}. \tag{2.11}$$

Moreover,

$$\mathrm{ES}(p) \iff (\{0\} \times \mathcal{W}_0(p)) \sqcup (\{1\} \times \mathcal{W}_1(p)) \neq \varnothing. \tag{2.12}$$

Proof. The last coordinate in (2.10) is integral. To see this without assuming it, reduce the second identity in (2.9) modulo u, multiply by a_j, and use $u \mid a_j^2$: this gives $u \mid p^\varepsilon a_j K$. The two identities (2.11) now follow from the first identity in (2.9). Their product is N_j^2, so Proposition 1.2 shows that (2.10) lies in $\mathscr{S}(p)$.

Conversely, suppose $\mathscr{S}(p) \neq \varnothing$, and sort a solution. Lemma 1.1 shows that its smallest denominator is uniquely of the form $x = a_j$ with $j \in J_p$: explicitly

$$R_j = 4x - p \equiv 3 \pmod 4, \qquad j = \frac{4x - p - 3}{4}.$$

Proposition 1.2 supplies a divisor $d \in \mathscr{D}_{p,j}$, and Proposition 1.3 places either d or its complement in the image of ι_E, or places d in the image of ι_M. Hence one of the two marked selector sets is nonempty. The forward implication follows from (2.10). □

1.3 The exact selector–terminal-lattice equivalence

Define the normalized, positively oriented, divisor-supported terminal slice

$$\mathcal{H}_\varepsilon(p) = \left\{ (j,\ell_0,\ell_1) \in \mathbb{Z}^3 : \begin{array}{l} j \in J_p, \quad \lambda(\ell_0,\ell_1,1) = p^\varepsilon, \\ T(\ell_0) := 4\ell_0 - 1 > 0, \\ u(j,\ell_0,\ell_1) := \ell_1 + jT(\ell_0) > 0, \\ u(j,\ell_0,\ell_1) \mid a_j^2 \end{array} \right\}. \tag{3.1}$$

Let $\mathcal{H}_\varepsilon(p,j)$ be the fibre with the displayed marker j.

For a selector state define its relation matrix

$$\mathsf{A}_\varepsilon(p;j,u) = \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix}, \qquad R_jT = 4u + p^\varepsilon. \tag{3.2}$$

For the marker j, retain both inverse unimodular shears

$$\mathsf{U}_j = \begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{L}_j = \mathsf{U}_j^{-1} = \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}. \tag{3.3}$$

Theorem 1.5 (strong marked bridge theorem)**.** *For every $\varepsilon \in \{0,1\}$, the map*

$$\Theta_{\varepsilon,p} : \mathcal{W}_\varepsilon(p) \longrightarrow \mathcal{H}_\varepsilon(p), \qquad (j,u) \longmapsto \left(j, \frac{T+1}{4}, u - jT\right) \tag{3.4}$$

is a bijection. Its inverse is

$$\Theta_{\varepsilon,p}^{-1}(j,\ell_0,\ell_1) = (j, \ell_1 + j(4\ell_0 - 1)). \tag{3.5}$$

For corresponding points, with $\ell = (\ell_0,\ell_1,1)$, one has the exact matrix identities

$$\boxed{\mathsf{U}_j\mathsf{A}_\varepsilon(p;j,u) = \mathsf{R}(\ell_0,\ell_1,1), \qquad \mathsf{A}_\varepsilon(p;j,u) = \mathsf{L}_j\mathsf{R}(\ell_0,\ell_1,1).} \tag{3.6}$$

In particular,

$$\mathrm{Rel}(\mathsf{A}_\varepsilon(p;j,u)) = \mathrm{Rel}(\mathsf{R}(\ell_0,\ell_1,1)), \qquad \det \mathsf{A}_\varepsilon = \det \mathsf{R} = -p^\varepsilon. \tag{3.7}$$

Thus the bridge intertwines the entire integral row presentation, not only its determinant.

Proof. Take $(j,u) \in \mathcal{W}_\varepsilon(p)$. Section 2 proves that T is a positive integer satisfying $T \equiv 3 \pmod 4$, so $\ell_0 = (T+1)/4 \in \mathbb{Z}$. Put $\ell_1 = u - jT$. Since $R_j = 4j+3$,

$$\begin{aligned} \lambda(\ell_0,\ell_1,1) &= 3(T+1) - 4(u - jT) - 3 \\ &= (4j+3)T - 4u = p^\varepsilon. \end{aligned}$$

The positivity and divisor-support conditions in (3.1) hold because $u = \ell_1 + jT$. Thus (3.4) is well-defined.

Conversely, take $(j,\ell_0,\ell_1) \in \mathcal{H}_\varepsilon(p)$, and set

$$T = 4\ell_0 - 1, \qquad u = \ell_1 + jT.$$

Expanding $\lambda(\ell_0,\ell_1,1) = p^\varepsilon$ gives

$$(4j+3)T - 4u = p^\varepsilon,$$

so $R_j \mid 4u + p^\varepsilon$. All other selector conditions occur explicitly in (3.1). The two constructions recover T, u, ℓ_0, ℓ_1 literally, which proves bijectivity and (3.5).

Finally, direct multiplication gives

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4j+3 & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -u+jT \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1-4\ell_0 \end{pmatrix}.$$

Both shears have determinant 1, so their integral row operations are invertible and the row lattices are equal. The signed determinant is $-R_jT + 4u = -p^\varepsilon$, proving (3.7). □

Corollary 1.6 (presented groups and the Erdős–Straus equation)**.** *For every corresponding selector/S6 point,*

$$\mathcal{G}(\mathsf{A}_\varepsilon(p; j, u)) = \mathcal{G}(\mathsf{R}(\ell_0, \ell_1, 1)) \cong G(\ell_0, \ell_1, 1) \cong \mathbb{Z}/p^\varepsilon\mathbb{Z}. \tag{3.8}$$

Here $\mathbb{Z}/1\mathbb{Z}$ is the trivial group. Moreover the direct map

$$\widehat{\Psi}_{\varepsilon,p} : \mathcal{H}_\varepsilon(p) \longrightarrow \mathscr{S}(p) \tag{3.9}$$

is obtained by (3.5), (2.8), and

$$(j, \ell_0, \ell_1) \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u}\right), \quad \begin{cases} T = 4\ell_0 - 1, \\ u = \ell_1 + jT, \\ K = (p^{1-\varepsilon}T + 1)/4. \end{cases} \tag{3.10}$$

It satisfies

$$\boxed{\mathrm{ES}(p) \Longleftrightarrow (\{0\} \times \mathcal{H}_0(p)) \sqcup (\{1\} \times \mathcal{H}_1(p)) \neq \varnothing.} \tag{3.11}$$

Proof. The relation $xy = c^{\ell_0}$ in (1.4) gives $y = x^{-1}c^{\ell_0}$; since c is central, x and y commute. The remaining two relations are the rows of (1.5). The greatest common divisor of the four entries of R is 1, because it divides both 3 and 4. Its Smith form is therefore $\mathrm{diag}(1, p^\varepsilon)$, proving (3.8). Formula (3.10) is the composition of Theorem 1.5 with Proposition 1.4, and (3.11) follows from (2.12). □

Remark 1.7 (the selector tag and the solution map). The channel tag is not redundant. For $p = 13$, $j = 0$, and $u = 2$, one has $u \in \mathcal{W}_0(p, j) \cap \mathcal{W}_1(p, j)$, but the two tags give different terminal values and different marked terminal points. Nor is the map to unordered denominator triples injective: in the M-channel at $p = 13, j = 0$, the complementary divisors $u = 2$ and $u = 8 = a_0^2/2$ both give the unordered solution $\{4, 26, 52\}$. The bijection in Theorem 1.5 is a bijection of marked selector states with marked terminal states, not a bijection with unordered solutions.

Corollary 1.8 (fixed-fibre affine coordinates and the intrinsic gate)**.** *Put*

$$C_\varepsilon = \frac{p^\varepsilon + 3}{4}, \qquad D_{\varepsilon,j} = j + C_\varepsilon = \frac{R_j + p^\varepsilon}{4}.$$

Every integral point with $\ell_2 = 1$ and $\lambda(\ell_0, \ell_1, 1) = p^\varepsilon$ is uniquely

$$(\ell_0, \ell_1, \ell_2) = (n, 3n - C_\varepsilon, 1), \qquad n \in \mathbb{Z}, \tag{3.12}$$

and, for each fixed marker $j \in J_p$, define

$$T = 4n - 1, \qquad u = R_j n - D_{\varepsilon,j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4}. \tag{3.13}$$

Then $T \equiv 3 \pmod 4$, so $K \in \mathbb{Z}$, and the congruence equation $R_j T = 4u + p^{\varepsilon}$ holds identically. On the locus $T > 0$ and $u > 0$, one has

$$\gcd(R_j, u) = 1, \qquad u \mid a_j^2 \Longleftrightarrow u \mid K^2. \tag{3.14}$$

Thus the nonlinear divisor gate can be written entirely in the S6 coordinates, but it does not disappear.

Proof. Equation (3.12) is the direct solution of $12\ell_0 - 4\ell_1 - 3 = p^{\varepsilon}$. Substitution gives the first two formulas in (3.13), and $T \equiv 3 \pmod 4$ proves that the displayed integer K is well defined. The definitions give

$$4u = 4R_j n - (R_j + p^{\varepsilon}) = R_j T - p^{\varepsilon},$$

so the congruence equation holds for every integral point in (3.12). If a positive integer divides both R_j and u, then the congruence equation makes it divide p^{ε}; (1.2) therefore gives $\gcd(R_j, u) = 1$. Without invoking the selector hypothesis, direct substitution gives

$$4\big(R_j K - p^{1-\varepsilon} u - a_j\big) = p^{1-\varepsilon}(R_j T - 4u) - p = 0.$$

Thus $a_j \equiv R_j K \pmod u$. Since R_j is a unit modulo u, $a_j^2 \equiv R_j^2 K^2 \pmod u$, which proves (3.14) in both directions. □

The normalization automatically lies in the admissible residue class used by the S6 terminal presentation:

$$\ell_2 = 1 \text{ is odd}, \qquad \ell_1 \equiv 2 \pmod 3. \tag{3.15}$$

Indeed reduction of $\lambda(\ell_0, \ell_1, 1) = p^{\varepsilon} \equiv 1 \pmod 3$ gives $-\ell_1 \equiv 1 \pmod 3$.

Remark 1.9 (orientation and the constructed S6 candidate). The geometrically reconstructed candidate elsewhere in the audited corpus uses the oriented terminal parameter $\lambda(0, 1, -1) = -1$. The two arithmetic channels in this note use the positive terminal values $+1$ and $+p$. No audited map identifies the -1 terminal datum with either positive datum. Consequently the row shear proved here does not, by itself, place any selector state on the particular complex threefold constructed in the -1 branch.

1.4 The exact remaining arithmetic incidence

The bridge can be expressed without an untyped phrase such as "some divisor of a square." The next factor coordinates are canonical and retain their orientation.

Lemma 1.10 (canonical ordered factor coordinates)**.** *For $a, u \in \mathbb{Z}_{>0}$, the condition $u \mid a^2$ is equivalent to the existence of a unique ordered triple $(r, s, t) \in \mathbb{Z}_{>0}^3$ such that*

$$u = r^2 s, \qquad a = rst, \qquad \gcd(r, t) = 1. \tag{4.1}$$

Explicitly, with $g = \gcd(a, u)$,

$$r = \frac{u}{g}, \qquad s = \frac{g^2}{u}, \qquad t = \frac{a}{g}. \tag{4.2}$$

Proof. Write $v_q(a) = \alpha$ and $v_q(u) = \beta \le 2\alpha$ prime by prime. Then

$$v_q(g) = \min(\alpha, \beta), \qquad v_q(g^2/u) = 2\min(\alpha, \beta) - \beta \ge 0.$$

Thus (4.2) consists of positive integers. If $\beta \le \alpha$, then $q \nmid r$; if $\beta \ge \alpha$, then $q \nmid t$. Hence $\gcd(r, t) = 1$, and direct multiplication gives (4.1).

Conversely, (4.1) gives $a^2/u = st^2 \in \mathbb{Z}_{>0}$. It also gives $\gcd(a, u) = rs$, because $\gcd(r, t) = 1$. Formula (4.2) then recovers (r, s, t), proving uniqueness. □

Proposition 1.11 (factor form of both selector channels)**.** *For the unique factorization $u = r^2 s$, $a_j = rst$, $\gcd(r,t) = 1$, one has*

$$u \in \mathcal{W}_0(p,j) \iff R_j \mid pr + t, \tag{4.3}$$

$$u \in \mathcal{W}_1(p,j) \iff R_j \mid r + t. \tag{4.4}$$

Proof. All of r, s, t are units modulo R_j, because their product is a_j and $\gcd(a_j, R_j) = 1$. For the E-channel,

$$t(4u+1) = 4r^2 st + t = (p + R_j)r + t = (pr + t) + R_j r.$$

Cancelling t modulo R_j proves (4.3). For the M-channel,

$$4u + p \equiv 4(u + a_j) = 4rs(r + t) \pmod{R_j},$$

and cancellation of the unit $4rs$ proves (4.4). □

Theorem 1.12 (necessary-and-sufficient counterexample criterion)**.** *A prime $p \equiv 1 \pmod{12}$ is a counterexample to the Erdős–Straus equation if and only if, for every $j \in J_p$ and every ordered $(r,s,t) \in \mathbb{Z}^3_{>0}$ satisfying*

$$a_j = rst, \qquad \gcd(r,t) = 1,$$

one has simultaneously

$$R_j \nmid pr + t, \qquad R_j \nmid r + t. \tag{4.5}$$

Equivalently, with the tag retained, it is necessary and sufficient that

$$R_j \nmid p^{1-\varepsilon} r + t \tag{4.6}$$

for every $j \in J_p$, every $\varepsilon \in \{0,1\}$, and every such factor triple.

Proof. Lemma 1.10 is a bijection between all allowed divisors in the selector (2.4) and all displayed ordered factor triples. Proposition 1.11 translates the two selector congruences without loss. The assertion now follows directly from (2.12), or equivalently (3.11), by negating the finite existential statement. □

This theorem isolates the unresolved problem exactly: the S6 row shear has not removed the bounded factor-incidence condition (4.5).

1.5 Why the labels and support cannot be forgotten

Lemma 1.13 (full terminal fibres are affine lattices)**.** *The kernel of $\lambda : \mathbb{Z}^3 \to \mathbb{Z}$ is*

$$\ker \lambda = \mathbb{Z}(1,3,0) \oplus \mathbb{Z}(1,0,4). \tag{5.1}$$

If $q \equiv 1 \pmod{12}$ and $m_q = (q+11)/12$, then

$$\lambda^{-1}(q) = \{(m_q + r + s,\ 2 + 3r,\ 1 + 4s) : r, s \in \mathbb{Z}\}. \tag{5.2}$$

In the normalized slice $\ell_2 = 1$, this is an infinite affine line.

Proof. If $12a - 4b - 3c = 0$, reduction modulo 4 gives $c = 4s$. The remaining equation gives $b = 3(a - s)$; putting $r = a - s$ yields $(a,b,c) = r(1,3,0) + s(1,0,4)$, uniquely. Since $\lambda(m_q, 2, 1) = 12m_q - 11 = q$, translation by (5.1) proves (5.2). □

The lemma proves that the mere existence of a terminal parameter with signed value 1 or p is automatic. It cannot imply selector nonemptiness. The following examples show separately why the marker and the divisor-support condition are both part of the theorem's domain.

Example 1.14 (forgetting j and p is noninjective)**.** For each eligible prime q, define the fixed-prime forgetful map

$$\mathfrak{f}_q : \mathcal{H}_0(q) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{f}_q(j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

At $p = 13$, the two E-states

$$(j, u, T) = (0, 2, 3), \qquad (1, 5, 3)$$

satisfy

$$\Theta_{0,13}(0, 2) = (0, 1, 2), \qquad \Theta_{0,13}(1, 5) = (1, 1, 2),$$

and both images are sent by $\mathfrak{f}_{13}$ to $(1, 2, 1)$. To compare different prime fibres, form the tagged coproduct and the global forgetful map

$$\mathfrak{F} : \bigsqcup_{\substack{q \text{ prime} \\ q \equiv 1 \pmod{12}}} (\{q\} \times \mathcal{H}_0(q)) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{F}(q; j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

The state $(p, j, u) = (73, 1, 5)$ has $\Theta_{0,73}(1, 5) = (1, 1, 2)$, so its tagged image also maps under $\mathfrak{F}$ to $(1, 2, 1)$. Therefore neither j nor the external prime p is recoverable after applying the explicitly displayed forgetful maps.

Example 1.15 (ambient positivity does not imply divisor support)**.** For $p = 13$, $j = 2$, one has $R_j = 11$ and $a_j = 6$. The positive normalized E-line has

$$(\ell_0, \ell_1) = (n, 3n - 1), \qquad u = 11n - 3,$$

and the positive normalized M-line has

$$(\ell_0, \ell_1) = (n, 3n - 4), \qquad u = 11n - 6.$$

Both lines are infinite. The only positive $u \le a_j^2 = 36$ are

$$E : \ 8, 19, 30, \qquad M : \ 5, 16, 27,$$

and none divides 36. Hence $\mathcal{H}_0(13, 2) = \mathcal{H}_1(13, 2) = \varnothing$, even though both ambient positive S6 lines are nonempty. The prime 13 is solved at other markers, so this is a fixed-shell counterexample to forgetting support, not an Erdős–Straus counterexample.

1.6 The natural unmarked S^6 point and its complete divisor gate

The most direct attempted use of the S6 presentation starts from one point in the signed p-fibre and shears it through every shell. Its exact content can be classified completely.

Write

$$p = 12m - 11, \qquad m = h + 1 \ge 2, \qquad J_p = \{0, \dots, 3m - 4\}. \tag{6.1}$$

Retain the unmarked S6 point

$$\ell^{\mathrm{aff}} = (m, 2, 1), \qquad \lambda(\ell^{\mathrm{aff}}) = p, \tag{6.2}$$

and for $j \in J_p$ put

$$T_0 = 4m - 1, \quad u_j = 2 + jT_0, \quad a_j = 3m + j - 2, \quad R_j = 4j + 3. \tag{6.3}$$

Then

$$R_j T_0 = 4u_j + p, \qquad R_j m = a_j + u_j, \qquad T_0 a_j - u_j = pm, \tag{6.4}$$

and

$$\mathsf{U}_j \begin{pmatrix} R_j & -u_j \\ 4 & -T_0 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 4 & 1 - 4m \end{pmatrix} = \mathsf{R}(m, 2, 1). \tag{6.5}$$

Theorem 1.16 (complete affine-line gate)**.** *For prime $p = 12m - 11$, $m \geq 2$, and $0 \leq j < 3m - 3$,*

$$u_j \mid a_j^2 \iff (j = 0 \text{ and } 2 \mid m) \text{ or } (j = 2 \text{ and } 8 \mid m). \tag{6.6}$$

Consequently the single unmarked point (6.2), with all markers restored, produces an Erdős–Straus witness if and only if $p \equiv 13 \pmod{24}$. In that case $j = 0, u = 2, K = m$ gives

$$\frac{4}{p} = \frac{1}{3m - 2} + \frac{1}{pm} + \frac{1}{pm(3m - 2)/2}. \tag{6.7}$$

For $p \equiv 1 \pmod{24}$, this affine S6 point supplies no supported selector state at any $j \in J_p$.

Proof. The identities (6.4) and (6.5) follow by direct expansion. Assume first that $u_j \mid a_j^2$, and put $g = \gcd(u_j, a_j)$. Write $u_j = gu'$ and $a_j = ga'$ with $\gcd(u', a') = 1$. The assumed divisibility gives $u' \mid g(a')^2$, hence $u' \mid g$, and therefore

$$u_j \mid g^2. \tag{6.8}$$

The last identity in (6.4) gives $g \mid pm$. Since $0 < a_j < p$, the primality of p and $g \mid a_j$ imply $p \nmid g$; therefore $g \mid m$. Reducing $u_j = 2 + j(4m - 1)$ modulo m now gives

$$g \mid j - 2. \tag{6.9}$$

Suppose $j \neq 0, 2$. Then $j \geq 1$, and (6.8)–(6.9) imply $u_j \leq g^2 \leq (j - 2)^2$. But

$$u_j - (j - 2)^2 = j(4m + 3 - j) - 2 > 0 \tag{6.10}$$

For $m \geq 2$ and $1 \leq j < 3m - 3$, indeed $4m + 3 - j \geq m + 7$, so the right-hand side of (6.10) is at least $m + 5 > 0$. This is a contradiction. Thus only $j = 0$ or $j = 2$ can occur.

At $j = 0$, $u_0 = 2$ and $a_0 = 3m - 2$, so $u_0 \mid a_0^2$ exactly when m is even. At $j = 2$, $u_2 = 8m$ and $a_2 = 3m$, so

$$u_2 \mid a_2^2 \iff 8m \mid 9m^2 \iff 8 \mid m.$$

These calculations also prove both converse implications in (6.6).

Now $p = 12m - 11$ is congruent to 13 (mod 24) exactly when m is even. The $j = 0$ state lies in the M-channel, and (2.8) gives $T = T_0$ and $K = (T_0 + 1)/4 = m$. Formula (2.10) is precisely (6.7). If m is odd, (6.6) excludes every marker on this particular S6 point. □

The last sentence is deliberately local to (6.2). For example, $p = 73$ has $m = 7$ odd, so the affine point contributes nothing; nevertheless $(j, u, T; \ell_0, \ell_1) = (1, 1, 11; 3, -10)$ is an M-state. Thus failure of the affine point is not failure of the full selector.

1.7 Exact descendant maps transported through the bridge

The bridge also transports a verified selector witness along multiplicative descendant rays. This is a map from an already nonempty domain, not a nonemptiness theorem.

Fix a supported state at (p, j), retain $R = R_j$, and let $b \in \mathbb{Z}_{>0}$. Define

$$a_b = ba_j, \qquad p_b = 4a_b - R = bp + (b-1)R. \tag{7.1}$$

Whenever p_b is again a prime congruent to 1 (mod 12), the same marker j belongs to J_{p_b}, since $R < p \le p_b$, and the following maps have the displayed domains and codomains:

$$\Phi^E_{b;p} : \mathcal{H}_0(p, j) \longrightarrow \mathcal{H}_0(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, \ell_0, \ell_1), \tag{7.2}$$

$$\Phi^M_{b;p} : \mathcal{H}_1(p, j) \longrightarrow \mathcal{H}_1(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, b\ell_0, b\ell_1 - j(b-1)). \tag{7.3}$$

Write

$$\theta_{\varepsilon,p,j} : \mathcal{W}_\varepsilon(p, j) \longrightarrow \mathcal{H}_\varepsilon(p, j), \qquad \theta_{\varepsilon,p,j}(u) = \Theta_{\varepsilon,p}(j, u)$$

for the fixed-marker restriction of the marked bridge.

Proposition 1.17 (intertwining of descendant actions)**.** *The selector maps corresponding to (7.2) and (7.3) are the typed maps*

$$\psi^E_{b;p} : \mathcal{W}_0(p, j) \longrightarrow \mathcal{W}_0(p_b, j), \qquad u \longmapsto u,$$

$$\psi^M_{b;p} : \mathcal{W}_1(p, j) \longrightarrow \mathcal{W}_1(p_b, j), \qquad u \longmapsto bu.$$

They intertwine the fixed-marker bridges exactly:

$$\Phi^\bullet_{b;p} \circ \theta_{\varepsilon,p,j} = \theta_{\varepsilon,p_b,j} \circ \psi^\bullet_{b;p}, \qquad (\bullet, \varepsilon) \in \{(E, 0), (M, 1)\}. \tag{7.4}$$

Equivalently,

$$\theta^{-1}_{0,p_b,j} \Phi^E_{b;p} \theta_{0,p,j}(u) = u, \qquad \theta^{-1}_{1,p_b,j} \Phi^M_{b;p} \theta_{1,p,j}(u) = bu.$$

On the accompanying arithmetic metadata these maps replace (a_j, p, u) by (a_b, p_b, u) and (a_b, p_b, bu), respectively. For the latter,

$$T_b = bT + b - 1, \qquad u_b = bu, \qquad K_b = bK, \tag{7.5}$$

and

$$\lambda(b\ell_0, b\ell_1 - j(b-1), 1) = p_b. \tag{7.6}$$

For either tag the maps are injective. For $c \in \mathbb{Z}_{>0}$, put

$$a_{bc} = bc\, a_j, \qquad p_{bc} = 4a_{bc} - R = bc\, p + (bc-1)R.$$

Whenever p_b and p_{bc} are primes congruent to 1 (mod 12)*, the displayed maps are composable and satisfy*

$$\Phi^\bullet_{c;p_b} \circ \Phi^\bullet_{b;p} = \Phi^\bullet_{bc;p} \tag{7.7}$$

Proof. In the E-channel, $u \mid a_j^2$ implies $u \mid a_b^2$, and the congruence $R \mid 4u + 1$ and quotient T are unchanged. Hence ℓ_0, ℓ_1 are unchanged by (3.4).

In the M-channel, $bu \mid b^2 a_j^2 = a_b^2$, and

$$4bu + p_b = b(4u + p) + (b-1)R.$$

Thus the descendant is a selector state and its quotient is $T_b = bT + b - 1$. Since $K = (T+1)/4$ in this channel, $K_b = (T_b + 1)/4 = bK$. Applying (3.4) gives

$$\ell_{0,b} = \frac{T_b + 1}{4} = b\ell_0, \qquad \ell_{1,b} = bu - jT_b = b\ell_1 - j(b-1),$$

which proves (7.3)–(7.6), including the intertwining identity (7.4). Each map is visibly injective for fixed b. Substitution into (7.1)–(7.3), with the source prime retained, proves the typed composition law (7.7). □

Reverse descent need not preserve divisor support: the factor b can introduce divisors absent from a proposed parent. Therefore (7.2)–(7.7) propagate witnesses but cannot create the first witness on an empty ray.

1.8 Sharp verdict

The strongest theorem supported by the audited data is Theorem 1.5 together with the exhaustive equivalence (3.11): the complete tagged Erdős–Straus selector for a fixed prime $p \equiv 1 \pmod{12}$ is exactly the marked divisor-supported S6 terminal slice, and the two integral relation matrices have the same row lattice. The exact remaining obstruction is (4.5).

Three stronger readings are false or unsupported.

1. The abstract group $G(\ell)$, its determinant, or an unmarked terminal triple does not determine selector support. Every E-state has the same trivial presented group, every M-state has group $\mathbb{Z}/p\mathbb{Z}$, and Examples 1.14 and 1.15 exhibit the information lost by forgetting the marker and divisor gate.

2. The row shear is not a geometric map from the reciprocal cubic $4xyz = p(xy + xz + yz)$ to the candidate complex threefold X. No map of varieties or analytic spaces, no cycle map, no attaching map, and no cohomological correspondence is defined by left multiplication of a relation matrix.

3. The existence of the full S6 affine fibre cannot prove universal Erdős–Straus positivity. Those fibres are always nonempty by (5.2), whereas selector nonemptiness is exactly the additional finite nonlinear incidence in (4.5). The natural affine point is completely exhausted by Theorem 1.16 and contributes nothing to the unresolved $p \equiv 1 \pmod{24}$ half.

Accordingly, the bridge is a genuine theorem and is stronger than a shared determinant or a heuristic analogy. It is also exact enough to prove its own boundary: without a new argument forcing one of the bounded factor congruences in (4.5), it neither proves nor disproves the Erdős–Straus conjecture.